

PSMFlows: ZERO-SHOT REINFORCEMENT LEARNING ON LIMITED-SUPPORT DATA VIA FLOW-INDEXED SUCCESSOR MEASURES

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ABSTRACT

Zero-shot reinforcement learning methods such as Forward–Backward representations (FB) and Proto Successor Measures (PSM) learn task-agnostic occupancy representations from reward-free data, but their training loops evaluate candidate actions that the dataset never supports: value estimates at out-of-distribution actions are unconstrained, and the policy-improvement argmax seeks out exactly those errors. We propose *PSMFlows*, which removes this failure mode by construction. A behavior-cloned conditional flow $G_\theta(s, u)$ maps latent noise to data-like actions; holding the noise fixed defines a family of policies $\pi_u(s) = G_\theta(s, u)$ indexed by u , and a successor-measure representation $\psi(s, u', u)^\top \varphi(x)$ is learned over this family, with dataset transitions carrying their own latents via flow inversion. Every action the method considers—during training and at inference—is a decode of the frozen flow, converting the hard constraint $a \in \text{supp } \mu_{\mathcal{D}}(\cdot | s)$ into the benign constraint that $\|u\|$ be typical. We describe the three-stage pipeline, a staged diagnostic protocol that makes each assumption falsifiable in isolation, and report first results on OGBench: the flow, inversion, and latent-family stages certify cleanly, while first representation-learning runs isolate a concrete failure mode—the learned measure discriminates policy indices without aligning them to realized behavior—that our diagnostics localize and that motivates the next iteration of the training loss.

1 INTRODUCTION

Behavior foundation models promise a single reward-free pretraining phase after which any downstream reward can be solved zero-shot. The dominant family—FB (Touati & Ollivier, 2021) and PSM (Agarwal et al., 2024)—learns linear-algebraic representations of successor measures. Applied to *offline* data with limited support, however, these methods inherit the classic offline-RL pathology: their Bellman maximization queries actions the dataset cannot certify. Conservatism penalties (VC-FB; Jeon et al., 2024) treat the symptom in the loss; we instead change the policy space so the symptom cannot arise.

Contributions. (i) a formulation of zero-shot offline RL as successor-measure learning over a *flow-indexed* policy family, in which the training distribution has behavior-policy concentrability $C = 1$; (ii) an in-sample training algorithm in which dataset transitions carry their own latents via flow inversion, with a relaxed-posterior (Laplace + importance-sampled EM) preimage operator; (iii) a staged, falsifiable diagnostic protocol (D1–D4) that separates flow fidelity, index coherence, inversion health, and control; (iv) an empirical study on OGBench, including a negative result whose diagnosis is, we argue, itself informative about what successor-measure losses ground and what they do not.

2 PRELIMINARIES

Setting. A reward-free MDP $(\mathcal{S}, \mathcal{A}, P, \gamma)$ with $\mathcal{A} \subseteq \mathbb{R}^{d_a}$; an offline dataset $\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^N$ collected by an unknown behavior policy $\mu(\cdot | s)$ with state marginal ρ . At test time a reward r is revealed (as labels on a sample of $\mathcal{D}$) and a policy must be produced with no further interaction.

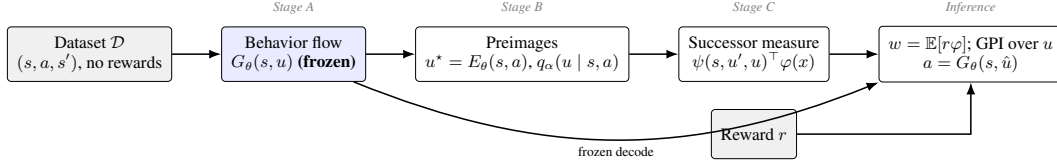


Figure 1: **The PSMFlows pipeline.** A reward-free behavior flow (Stage A, frozen) turns noise into data-like actions; flow inversion (Stage B) equips every dataset transition with its latent; a successor measure over the flow-indexed policy family is trained in-sample (Stage C); at test time a reward is projected onto the state basis and generalized policy improvement runs entirely in latent space. An editable draw.io source of this figure is provided in `figures/pipeline.drawio`.

Successor measures. For a policy π , $M^\pi(s, a, X) = \mathbb{E}[\sum_{t \geq 1} \gamma^{t-1} \mathbf{1}\{s_t \in X\} \mid s_0 = s, a_0 = a]$ is the discounted occupancy of $X \subseteq \mathcal{S}$. Then $Q_r^\pi(s, a) = \int r dM^\pi(s, a, \cdot)$: one object serves every reward, which is the basis of zero-shot RL. FB parameterizes $m^\pi(s, a, x) = F(s, a, z_\pi)^\top B(x)$; PSM uses an affine decomposition $m^\pi = b + \Phi w_\pi$ whose policy dependence is linear. Both train by a measure-matching temporal-difference loss and infer tasks by projecting r on the state basis.

The limited-support problem. The TD loss and the actor of these methods evaluate Q at actions from a training policy π_{tr} . The relevant quantity is the concentrability $C(\pi_{\text{tr}}) = \text{ess sup } \pi_{\text{tr}}(a \mid s) / \mu(a \mid s)$: FB’s deterministic actor gives $C = \infty$; PSM’s uniform action sampling gives C finite only for full-support behavior. On play or mode-restricted data neither holds, and value errors at unsupported actions are exactly the ones the argmax selects.

Conditional flow matching. A velocity field $v_\theta(s, u_t, t)$ trained with the CFM regression $\mathbb{E}\|v_\theta(s, (1-t)u_0 + ta, t) - (a - u_0)\|^2$, $u_0 \sim \mathcal{N}(0, I)$, induces a decoder $G_\theta(s, \cdot)$: the time-one map of the ODE $\dot{u} = v_\theta(s, u, t)$. With latent dimension equal to action dimension, $G_\theta(s, \cdot)$ is a C^1 diffeomorphism of $\mathbb{R}^{d_a}$; its inverse $E_\theta(s, \cdot)$ is computed by backward integration.

3 METHOD: PSMFLOWS

3.1 A FLOW-INDEXED POLICY FAMILY

Fix the trained (and thereafter frozen) behavior flow. For $u \in \mathbb{R}^{d_a}$ define the fixed-index policy $\pi_u(s) = G_\theta(s, u)$. Two properties make this family the right search space for offline zero-shot RL. First, *mass conversion*: for any set U of prior mass $1 - \delta$, the decoded actions carry $1 - \delta$ of the behavior mass *simultaneously at every state*, so “stay on the data” becomes “keep $\|u\|$ typical”. Second, *divergence isometry*: for any ν over latents, $D_f(\pi_\nu \parallel \hat{\mu}_\theta) = D_f(\nu \parallel p_0)$ at every state, so trust-region constraints against the (unknown) behavior density become closed-form constraints against a standard normal.

3.2 DATASET LATENTS BY FLOW INVERSION

Training a measure over $\{\pi_u\}$ requires transitions labeled with latents. Since $G_\theta(s, \cdot)$ is invertible, each dataset action has a point preimage $u_i^* = E_\theta(s_i, a_i)$; because a whole basin of latents decodes near a_i , we also fit the relaxed posterior

$$q_\alpha(u \mid s, a) \propto \mathcal{N}(u; 0, I) \exp(-\alpha \|G_\theta(s, u) - a\|), \quad (1)$$

by importance-sampled EM initialized from the Laplace approximation with covariance $\propto (J^\top J)^{-1}$, $J = \partial G_\theta / \partial u$ (Figure 2). The temperature α trades sharpness against effective sample size (ESS); both are measured and gated (§4). Under an exact flow, $E_\theta(s, a) \sim \mathcal{N}(0, I)$ on data, so the $\chi_{d_a}^2$ typicality of $\|u^*\|^2$ is a direct, falsifiable test of the flow-equals-behavior assumption.

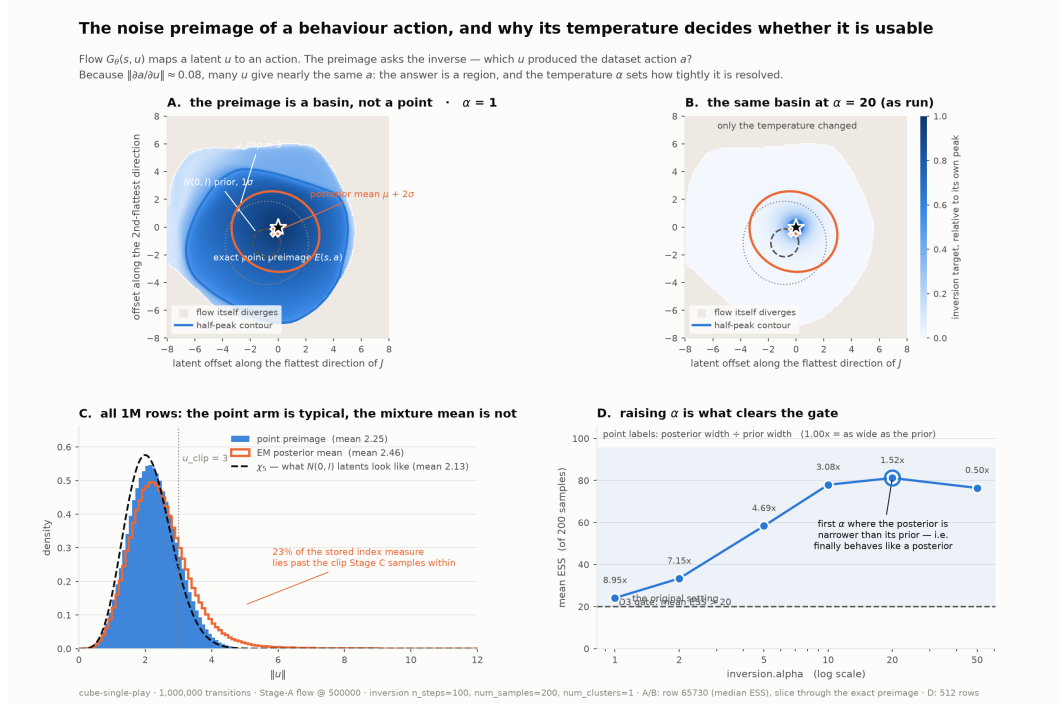


Figure 2: **What the noise preimage is, and why the temperature α decides whether it is usable.** (A) a 2D latent-space slice at one real cube transition: the preimage of an action is a *basin*, not a point; at $\alpha=1$ the inversion target (shading) is far broader than the $\mathcal{N}(0, I)$ prior (dashed). (B) the same slice at $\alpha=20$: the basin contracts to prior scale. (C) aggregate over 10^6 transitions: point preimages track the χ_{da} typical radius, while low- α posterior means drift far outside it. (D) mean ESS versus α against the D3 gate.

3.3 SUCCESSOR MEASURES OVER THE FAMILY

We learn $m(s, u, u', x) = \psi(s, u', u)^\top \varphi(x)$, the density of $M^{u \rightarrow u'}(s, \cdot)$: take the action indexed by u now, then follow $\pi_{u'}$. The Bellman identity closes on the diagonal, $M^{u \rightarrow u'}(s, X) = \mathbb{E}_{s'}[\mathbf{1}\{s' \in X\} + \gamma M^{u' \rightarrow u'}(s', X)]$, so a single network suffices. The empirical loss is the standard measure-matching TD objective with batch negatives, an orthonormality regularizer on φ , and three sampling rules: the action slot u is drawn from the stored preimage posterior of the transition (clipped to the typical set); the index slot u' is drawn fresh (mixed $\mathcal{N}(0, I)$ /permuted behavior latents); the bootstrap target is $\bar{\psi}(s', u', u')$ —the continuation latent *is* the policy index. No decoded action appears anywhere in training; under an exact flow the training distribution has concentrability $C = 1$ w.r.t. the behavior policy.

3.4 INFERENCE

Given rewards on relabeled data, $w = \frac{1}{N} \sum_i r(s'_i) \varphi(s'_i)$ is the least-squares task projection (exact under orthonormal φ), and $\psi(s, u', u)^\top w$ estimates the corresponding Q . Sampled GPI over the fixed-index family draws K candidate latents and M indices, takes $\hat{u} = \arg \max_k \max_m \psi(s, u'_m, u_k)^\top w$, and acts $a = G_\theta(s, \hat{u})$. Section 4 shows this is bounded by what the family contains, which motivates the variant below.

3.5 LATENTFLOWPSM

The support construction does not require that a *fixed* latent be the policy—only that every executed action be a flow decode. LatentFlowPSM keeps the frozen decoder and moves policy identity into

the task vector. Learned are features $\varphi(x)$, successor features $\psi(s, u, w)$ with $m(s, u, w, x) = \psi(s, u, w)^\top \varphi(x)$ and $Q_w(s, u) = \psi(s, u, w)^\top w$, an actor $\pi_\eta(s, w, \epsilon)$ emitting a latent in the clip box, and a flow-matching field over dataset latents used as the actor’s behavior-cloning anchor. The measure loss is the usual measure-matching objective with one change: the continuation latent at s' is the actor’s, $u^+ = \pi_\eta(s', w, \epsilon)$, rather than a fixed index, so policy improvement happens in latent space while every bootstrapped action remains a decode. The actor is trained by deterministic policy gradient on Q_w with the flow-matching anchor and ensemble pessimism; ψ takes no actor gradient and G_θ takes none at all. At deployment $w = \mathbb{E}_{\mathcal{D}}[r\varphi]$ and $a = G_\theta(s, \pi_\eta(s, w, \epsilon))$, with no test-time training.

4 EXPERIMENTS AND CURRENT FINDINGS

Protocol. OGBench (Park et al., 2024) pointmaze-medium-navigate and cube-single-play (state-based, single-task rewards); FQL (Park et al., 2025) as the per-task topline; in-repo, parity-verified PSM and FB as zero-shot baselines, and antmaze-medium-navigate for the latent-actor results. Headline numbers are 500-episode evaluations of the final checkpoint with Wilson 95% intervals; across-seed summaries are mean $\pm$ 95% CI over ≥ 3 seeds. Every zero-shot number is quoted beside the per-step-prior behavior-cloning control decoded from the same frozen flow, since that is the baseline the method must beat. Gates D1–D4 run in order, which localizes any failure to a single stage.

Gate	Question	Result (pointmaze / cube)	Status
D1	flow reproduces data actions	MMD $3 \cdot 10^{-4} / 1.5 \cdot 10^{-4}$ (ctrl. $\times 13 / \times 10^3$)	pass
D2	fixed u = distinct, repeatable policy	consistency ratio 0.53 / 0.51	pass [†]
D3	inversion trustworthy	ESS 91 / 82; roundtrip $\sim 10^{-4}$; typicality ≥ 0.99	pass
D4	oracle-reward control $\geq 50\%$ of FQL	0.00 vs. FQL peak 0.687 ± 0.201	fail

Table 1: **Staged diagnostics (current state).** The generative substrate certifies cleanly; control does not yet. The point of the protocol is that the failure is localized: it lies in the policy family itself, not the flow or the inversion. [†]D2 measures whether fixed- u policies are repeatable and mutually distinct. They are; that is not the same as being *useful*, and §4 shows the gate cannot see the difference.

Finding 1: the substrate works. The behavior flow beats random controls by 1–3 orders of magnitude on distribution match (D1); fixed-latent policies are reproducible and mutually distinct (D2); and at $\alpha=20$, $N=200$ the preimage posterior passes every inversion gate (D3), with 99% of inverted latents inside the central $\chi^2_{d_a}$ typicality band—direct evidence for the exact-flow assumption at the fidelity our gates demand.

Finding 2: the temperature is measurable, and it matters. At $\alpha=1$ the Laplace proposal collapses to the prior (99.8% of rows fully clipped), 39% of posterior means leave the typical set, and ESS ≈ 10 ; at $\alpha=20$ posterior means track the χ_{d_a} radius and ESS exceeds the gate five-fold (Figure 2). Latents computed at the wrong temperature are quantifiably noise.

Finding 3 (negative, root-caused): the fixed-latent family has no goal-reaching member. Representation runs over the fixed-index family reach 0.00 success on pointmaze and 0.02–0.06 on cube even with oracle rewards. This is not a training failure: it is a property of the family. Rolling *every* latent on a grid that covers the whole box—plus dataset preimages and the preimages of goal-reaching transitions—for full episodes, 0 of 233 candidates ever reach the pointmaze goal, 2 of 128 on cube, 5 of 64 on antmaze (Figure 4). The mechanism is temporal, not per-state: at a fixed state the decoded action covers every direction (angular circular variance 0.45–0.78), but within an episode the inverted latents are white (lag-1 autocorrelation 0.25, decaying to zero by lag 50; within-episode variance is 0.99 of the marginal). The behavior flow factorizes behavior into a state-conditional and a per-step quantile, and the expert’s route depends on a goal that is not in the observation, so route identity is pushed into the latent *sequence* and never into any single u . Consequently every π_u has the same value under any goal-reaching reward, $\psi^\top w$ is flat over candidates ($\pm 2.5\%$), and GPI’s argmax drifts to the saturated corners. The prediction this makes is quantitative and it holds: flow-GPI over fixed u scores 0.015 on cube against a measured family ceiling of $2/128 = 0.0156$. GPI was already extracting everything the family contained.

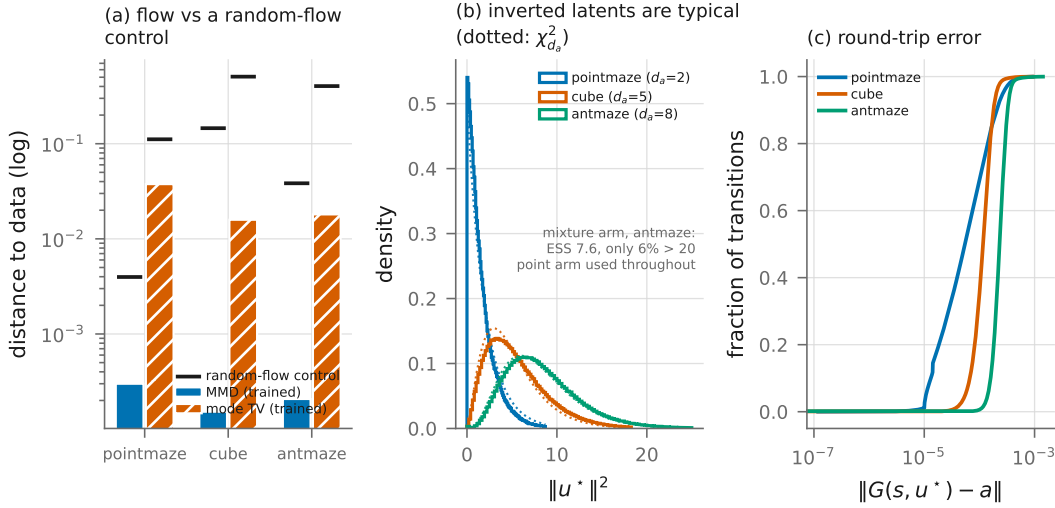


Figure 3: **The generative substrate certifies.** (a) The trained flow is one to three orders of magnitude closer to the behavior distribution than a random-flow control on both metrics. (b) Inverted latents follow the $\chi_{d_a}^2$ law they should follow under the prior in all three environments. The mixture arm is the one exception and is labeled as such: at $d_a=8$ its ESS collapses, so the point preimage is used throughout. (c) Decoding the recovered latent reproduces the recorded action to $\sim 10^{-4}$. Whatever fails downstream, it is not inversion quality.

Finding 4: moving the policy index into a task vector recovers most of the loss. The support construction survives Finding 3—every action is still a flow decode—but the premise that a fixed noise vector *is* a policy does not. We therefore keep the frozen flow and make the index a task vector w , with an amortized actor $u_\eta(s, w)$ emitting a latent per step and the measure backup bootstrapping the actor’s latent at s' rather than a fixed index (*LatentFlowPSM*, §3). On cube-single this reaches 0.236 success (500 episodes/seed, 5 seeds, Wilson [0.219, 0.253]; across seeds 0.236 ± 0.071) against a behavior-cloning control of 0.068 [0.049, 0.093] measured from the same frozen flow—a $3.5\times$ improvement with non-overlapping intervals (Figure 5). On antmaze-medium it reaches 0.222 [0.188, 0.261] against a control of 0.090 [0.068, 0.118]. Two results temper this. First, the actor-free ablation—the same representations acting by per-step latent argmax—scores 0.055 [0.047, 0.064], *below* the behavior-cloning control: ψ is a usable training signal for the actor but not a reliable ranker of latents, which is the same weakness Finding 3 diagnosed, surviving the redesign. Second, per-task FQL reaches 0.949 [0.936, 0.960] on the same task, so LatentFlowPSM recovers about a quarter of what a reward-supervised learner obtains, and its curve is flat from 50k steps—it converges to that level rather than being budget-limited.

Finding 5 (open): the measure still does not rank. Recording 60 episodes per seed of the deployed policy (Figure 6) settles the support claim and sharpens the open problem. The latents the actor emits are comfortably inside the prior’s typical set—mean $\|u\|^2 = 3.87$ on cube against $d_a=5$, 5.64 on antmaze against $d_a=8$, narrower than even the dataset latents (5.59, 8.49), with no mass on the clip boundary—so every executed action is a decode of a typical latent, as constructed. But the value the actor was trained against does not predict what happens: the predicted $\psi(s, w, u)^\top w$ at the initial state separates successful from failed episodes with AUC 0.588 [0.480, 0.694] on cube (180 episodes, 33 successes) and 0.354 [0.168, 0.567] on antmaze, both intervals covering chance. This is the same defect the GPI ablation exposes, measured a second way: $\psi^\top w$ is a serviceable gradient signal for the actor and an unreliable estimate of value. Why deterministic policy gradient on an uninformative critic nonetheless produces a policy that beats its behavior prior by $3.5\times$ is, as far as we can tell, unexplained, and is the first thing the next round of work should isolate.

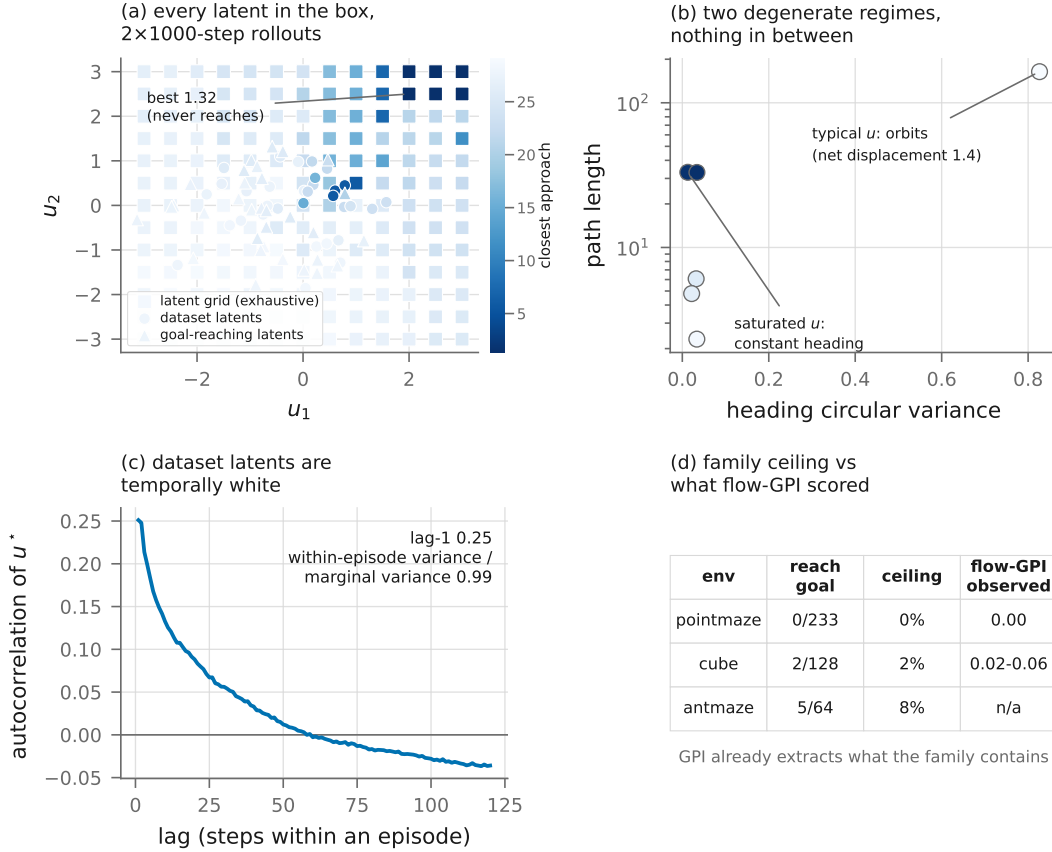


Figure 4: **The fixed-latent family contains no goal-reaching policy.** (a) With $d_a=2$ a 13×13 grid covers the *entire* latent box, so this is the whole family, not a sample: none of the 233 candidates reaches the goal in 2×1000 steps and the best approach is 1.32 in a maze ~ 30 across. (b) The family collapses into two regimes: saturated latents give a near-constant heading, typical latents give a field that circulates (path length 164, net displacement 1.4). (c) Dataset latents are temporally white, so no fixed u encodes a route. (d) What flow-GPI scored equals what the family contains.

5 RELATED WORK

Deferred to the next draft; see RESEARCH_NOTE §6 for the verified map.

6 CONCLUSION

PSMFlows converts the support constraint of offline zero-shot RL into latent-space typicality by construction. Its generative substrate is validated; its open problem—grounding the policy-index axis of the successor measure against realized behavior—is now precisely characterized, and is the subject of ongoing work.

REPRODUCIBILITY STATEMENT

All gates, tools, and launchers are scripted; every diagnostic writes a JSON artifact; preimage archives carry sidecars recording the exact checkpoint and inversion settings that produced them.

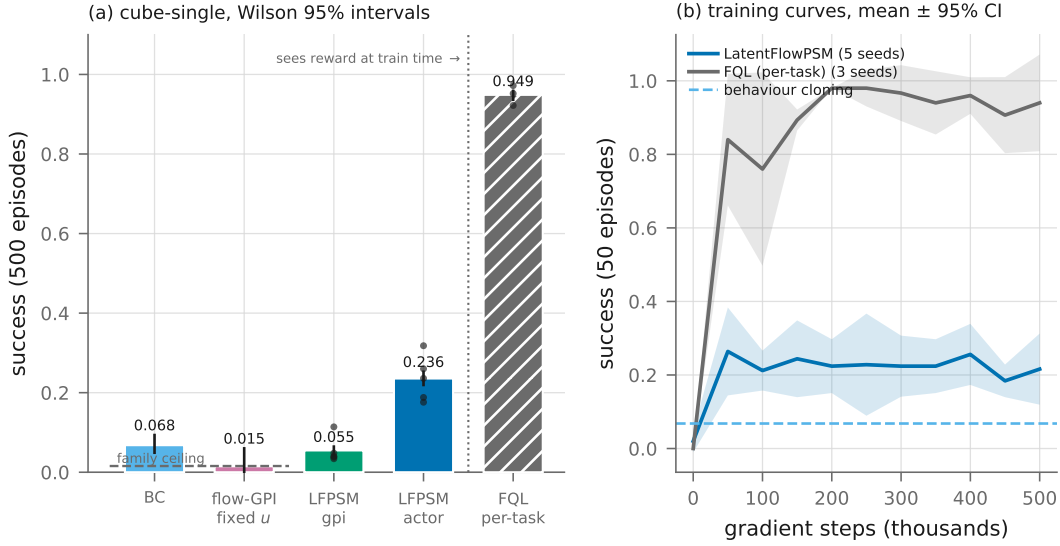


Figure 5: **Which way of choosing a latent earns its keep** (cube-single, 500-episode evaluations, Wilson 95% intervals, per-seed dots). Selection over fixed latents sits at the family ceiling measured in Figure 4. The amortized latent actor clears the behavior-cloning control by $3.5\times$; the actor-free ablation on identical representations falls below it. FQL is separated because it sees the reward during training: a ceiling, not a peer.

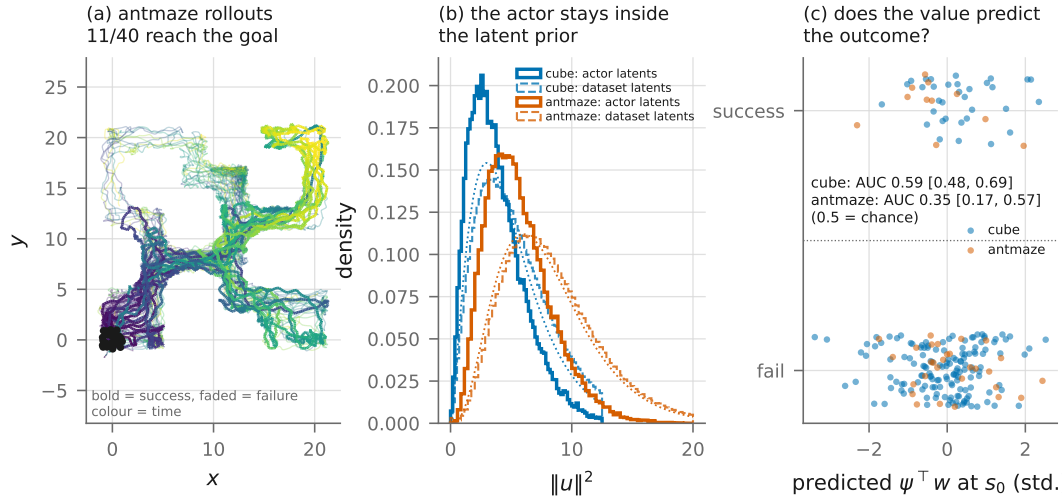


Figure 6: **What the deployed policy does.** (a) Antmaze rollouts, colored by time. (b) The latents the actor emits stay inside the prior’s typical set, beside the dataset latents and the $\chi^2_{d_a}$ density—the in-support claim, measured rather than asserted. (c) Predicted $\psi(s, w, u)^T w$ at the initial state against the outcome of that episode, as AUC against binary success (failed episodes all return the same value, so a rank correlation over that tie mass is uninterpretable). Both intervals cover chance.

REFERENCES

Siddhant Agarwal, Ishan Durugkar, Peter Stone, and Amy Zhang. Proto successor measure: Representing the space of all possible solutions of reinforcement learning. *arXiv preprint arXiv:2411.19418*, 2024.

Scott Jeen, Tom Bewley, and Jonathan M Cullen. Zero-shot reinforcement learning from low quality data. *Advances in Neural Information Processing Systems*, 37, 2024.

Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. Ogbench: Benchmarking offline goal-conditioned rl. *arXiv preprint arXiv:2410.20092*, 2024.

Seohong Park, Qiyang Li, and Sergey Levine. Flow q-learning. *arXiv preprint arXiv:2502.02538*, 2025.

Ahmed Touati and Yann Ollivier. Learning one representation to optimize all rewards. *Advances in Neural Information Processing Systems*, 34, 2021.

A APPENDIX